PG No. 7 C_{2v} $mm2$ [orthorhombic]

Table 1: Representation matrices

Irrep.				
A_1	$1: \begin{bmatrix}1\end{bmatrix}$	$2_{001}: \begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}: \begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}: \begin{bmatrix}1\end{bmatrix}$
A_2	$1: \begin{bmatrix}1\end{bmatrix}$	$2_{001}: \begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}: \begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}: \begin{bmatrix}-1\end{bmatrix}$
B_1	$1: \begin{bmatrix}1\end{bmatrix}$	$2_{001}: \begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}: \begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}: \begin{bmatrix}-1\end{bmatrix}$
B_2	$1: \begin{bmatrix}1\end{bmatrix}$	$2_{001}: \begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}: \begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}: \begin{bmatrix}1\end{bmatrix}$